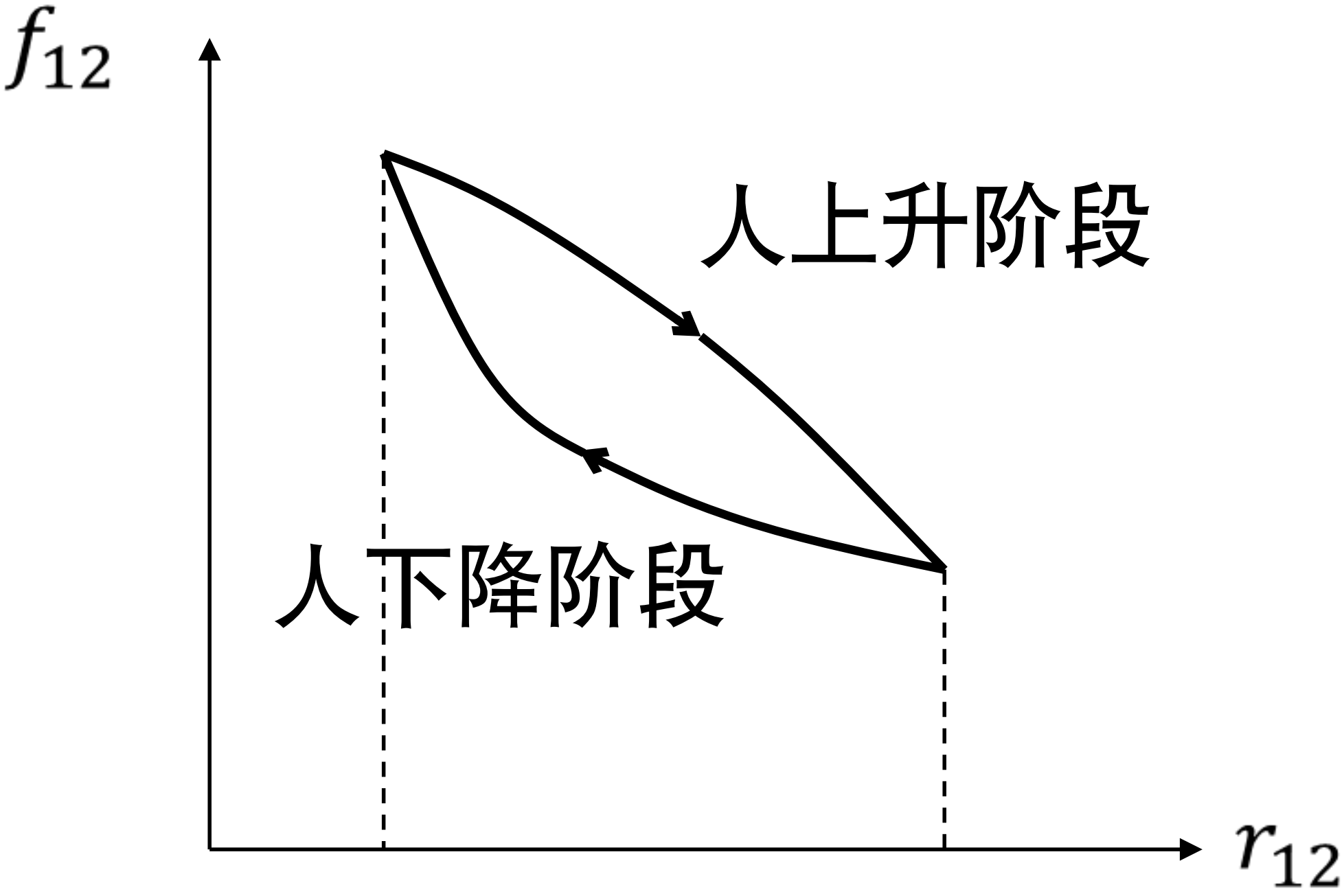
f_{12}
人上升阶段
人下降阶段
r_{12}